

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}$$

$$\Leftrightarrow \sum_{j=1}^n a_{ij} x_j = b_i \quad i=1, \dots, m$$

$m = \text{numero eq.} = \text{numero incognite}$

$$A = (a_{ij})_{i,j=1..m} \in \mathbb{R}^{m \times n}$$

$$b = (b_i)_{i=1..m} \in \mathbb{R}^m$$



$$Ax = b$$

PAGE RANK $\underbrace{(I - \rho G D)}_A x = \underbrace{\delta e}_b$

$$\left[Ax = b \text{ ha soluzione} \Leftrightarrow \overset{r}{\text{rank}}(A) = \overset{r}{\text{rank}}([A|b]) \right]$$

$r = m \Rightarrow \exists! \text{ soluzione}$
 $r < m \Rightarrow \exists \infty \text{ soluzioni}$

TEOREMA DI CRAMER

$Ax = b$ quadrato
ammette una e una
soluzione

$$\Leftrightarrow \det(A) \neq 0$$

page Rank con $\rho = 1$

$$\underbrace{(I - G D)}_A x = \underbrace{0}_b \quad (*)$$

$$0 = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

Si dimostra che

$$\det(I - G D) \neq 0$$

CRAMER

$$\Rightarrow (*)$$

ha un unico vettore
x soluzione

$\det(I - G) \neq 0 \Rightarrow$ $\odot$ ha un'unica soluzione
 $\Rightarrow x=0$ è l'unica vettore soluzione:
 $A \cdot 0 = 0$

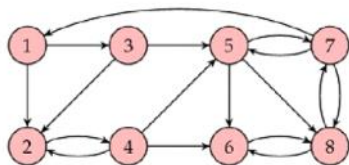
Command Window:

```
>> esercizioPageRank1
x =
    0.1318
    0.3329
    0.1878
    0.3475
```

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```
1 % A=[1 -p/3 0 0
2 % 0 1 0 -p
3 % -p/2 -p/3 1 0
4 % -p/2 -p/3 -p 1]
5 G=[0 1 0 0;
6 0 0 0 1;
7 1 1 0 0;
8 1 1 1 0];
9 n=size(G,1); p=0.85; delta=(1-p)/n;
10 I=eye(n);
11 c=sum(G,1);
12 D=diag(1./c);
13 e=ones(n,1);
14 A=I-p*G*D;
15 b=delta*e;
16 x=A\b
```

2. Dato il seguente grafo



deriva la matrice G di connettività e formula il problema del pagerank. Risolvi il problema utilizzando i codici Matlab a tua disposizione. Qual è il ranking delle pagine?

Command Window:

```
>> r=sum(G,2)
r =
    0.0631
    0.0925
    0.0456
    0.0974
```

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```
9 % 1 1 1 0];
10 % =====
11 % ESERCIZIO 2
12 G=[0 0 0 0 0 0 1 0
13 1 0 1 1 0 0 0 0
14 1 0 0 0 0 0 0 0
15 0 1 0 0 0 0 0 0
16 0 0 1 1 0 0 1 0
17 0 0 0 1 1 0 0 1
```

numero di link entranti:

1	0.0631
2	0.0925
3	0.0456
4	0.0974

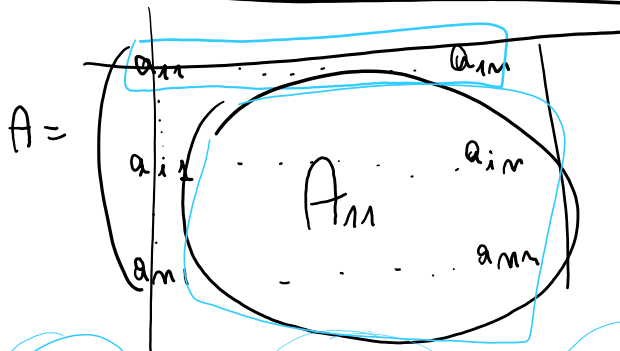
numero di
link
servienti:
in
ciascuna
pagina

ilue
8 dou
,12,1...
0188;0.0188;0.0188;0...
1,2,3,3,1,3,2]
8 double
1188
1;1;1;1;1;1]
8 double
R double

1 0.0925
3 0.0456
1 0.0974
1 0.1101
3 0.1841
3 0.1565
2 0.2508
3

fx >>

```
15 0 1 0 0 0 0 0 0
16 0 0 1 1 0 0 1 0
17 0 0 0 1 1 0 0 1
18 0 0 0 0 1 0 0 1
19 0 0 0 0 1 1 1 0];
20 n=size(G,1); p=0.85; delta=(1-p)/n;
21 I=eye(n);
22 c=sum(G,1);
23 D=diag(1./c);
24 e=ones(n,1);
25 A=I-p*G*D;
26 b=delta*e;
27 x=A\b
```



Rif $i: [a_{i1}, a_{i2}, \dots, a_{im}]$

$$\det(A) = +a_{11} \cdot \det(A_{11}) - a_{12} \det(A_{12}) + \dots + (-1)^{1+m} a_{1m} \det(A_{1m})$$

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(A_{ij})$$

matrice da cui si ottiene cancellando riga i e colonna j

Costo $n!$

INPUT A, b

OUTPUT x

Dimensione $n=3$

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{33}x_3 = b_3 \end{cases}$$

$a_{11} \neq 0, a_{22} \neq 0, a_{33} \neq 0$

costo: 6 M

$$\begin{cases} x_3 = \frac{b_3}{a_{33}} \\ x_2 = \frac{b_2 - a_{23}x_3}{a_{22}} \\ x_1 = \frac{b_1 - a_{12}x_2 - a_{13}x_3}{a_{11}} \end{cases}$$

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1m}x_m = b_1$$

$$\begin{aligned}
 a_{11}x_1 + a_{12}x_2 + \dots + a_{1m}x_m &= b_1 \\
 a_{22}x_2 + \dots + a_{2m}x_m &= b_2 \\
 &\vdots \\
 a_{ii}x_i + \dots + a_{im}x_m &= b_i \\
 &\vdots \\
 a_{mm}x_m &= b_m
 \end{aligned}$$

EQUAZIONE i

$$x_m = b_m / a_{mm}$$

per $i = m-1, \dots, 1$

$$x_i = \frac{b_i - \sum_{j=i+1}^m a_{ij}x_j}{a_{ii}}$$

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```

EDITOR      PUBLISH      VIEW
+ New      Open      Save      Find Files      Compare      Go To      Find      Insert      Comment      Breakpoints      Run      Run and Advance      Run Section      Run and Time
FILE      NAVIGATE      EDIT      BREAKPOINTS      RUN

esercizioBacks.m  backs.m
1-  U=[2 2 4
2-      0 7 11
3-      0 0 2];
4-  c=[5;8;2];
5-  x=backs(U,c)

1-  function x=backs(A,b)
2-
3-  n=length(b); %n=size(A,1) oppure n=size(A,2)
4-  for i=1:n
5-      if A(i,i)==0
6-          disp('Elemento nullo su diagonale!!!')
7-          return
8-      end
9-  end
10-
11-  x=zeros(n,1);
12-  x(n)=b(n)/A(n,n);
13-  for i=n-1:-1:1
14-      s=0;
15-      for j=i+1:n
16-          s=s+A(i,j)*x(j);
17-      end
18-      x(i)=(b(i)-s)/A(i,i);
19-  end

```

$$\begin{cases}
 2x_1 + 2x_2 + 4x_3 = 5 \\
 7x_2 + 11x_3 = 8 \\
 2x_3 = 2
 \end{cases}$$

$$\Rightarrow \begin{cases}
 x_3 = \frac{2}{2} = 1 \\
 x_2 = \frac{8 - 11x_3}{7} = \frac{8 - 11}{7} = -\frac{3}{7} \\
 x_1 = \frac{5 - 2x_2 - 4x_3}{2} = \frac{5 - 2(-3/7) - 4}{2} = \frac{13}{14}
 \end{cases}$$

```

Command Window
New to MATLAB? See resources for Getting Started.
>> esercizioBacks

x =

```

OK

```
>> esercizioBacks

x =

    0.9286
   -0.4286
    1.0000

>> -3/7

ans =

   -0.4286

>> 13/14

ans =

    0.9286
```

OK